Hypothesis two sample t-tests

1 In each of the following cases, test the hypothesis at the given level of significance and comment on whether the null hypothesis should be rejected or not rejected. Assume that all the populations are approximately normally distributed and have equal variance.

a $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 < \mu_2$, $\alpha = 10\%$
$\overline{x_1} = 411$, $n_1 = 130$, $S_{x_1} = 123$, $\overline{x_2} = 404$, $n_2 = 135$, $S_{x_2} = 52$.

b $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 > \mu_2$, $\alpha = 3\%$
$\overline{x_1} = 71$, $n_1 = 15$, $S_{x_1} = 17$, $\overline{x_2} = 75$, $n_2 = 13$, $S_{x_2} = 15$.

c $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$, $\alpha = 15\%$
$\overline{x_1} = 52.4$, $n_1 = 33$, $S_{x_1}^2 = 55$, $\overline{x_2} = 55.1$, $n_2 = 35$, $S_{x_2}^2 = 58$.

2 A researcher claims that on average Year 12 students study more than Year 11 students. He assumes the hours studied is normally distributed and decides to conduct a two sample t-test at a level of 1% significance to investigate. The researcher recorded the number of hours that two groups of Year 11 and 12 students studied over a week long period of time. For the group of 19 Year 11 students he found $\overline{x_1} = 11.2$ hrs per week with $S_{x_1} = 1.4$ hrs. For the group of 21 Year 12 students he found $\overline{x_2} = 13.1$ hrs per week with $S_{x_2} = 1.8$ hrs.

a Define the variables and hypothesis set for this problem.

b Is this a one tailed or two tailed test?

c Find the p-value for the hypothesis, correct to three significant figures.

d Comment on your results.

3 Buzz Electrics claim that the life spans of their candy bar vending machine light globes are normally distributed and that their new model bulbs last longer than their old model bulbs. A sample of 15 old model light globes are tested and found to have a mean of 1550 hours and a sample standard deviation of 80 hours. A sample of 18 new model light globes are tested and found to have a mean of 1600 hours and a sample standard deviation of 75 hours. They decide to conduct a two sample t-test to determine if the new light globes last longer than the old light globes using a significance level of 0.05.

a Define the variables and hypothesis set for this problem.

b Is this a one tailed or two tailed test?

c Find the p-value for the hypothesis, correct to two decimal places.

d Comment on your results.

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4 Red foxes are a species introduced to Australia that cause major damage to native wildlife. An environmental group wishes to use a two sample t-test to see if there is a difference between two of the leading bait poisons used to control the fox population numbers. They choose 50 unfenced plots of land and distribute baits with poison A on 25 plots and poison B on the other 25 plots. They record the number of dead foxes found on each of the plots and find that the mean number of deaths on plots using poison A is 5.2 foxes, with a standard deviation of 1.4. Plots using poison B have a mean of 4.6 foxes, with a standard deviation of 1.6.

a What two assumptions must the environmentalists make in order to use a two sample t-test?

b Define the variables and hypothesis set for this problem.

c Find the p-value for the hypothesis, correct to two decimal places.

d Comment on your results if the test is conducted at the 5% level of significance.

5 A pharmaceutical company has a new drug to help people sleep better. They want to find out if there is any improvement in sleep outcomes of people taking the new drug. As the data is normally distributed they conduct a two sample t-test experiment. They first give all people a placebo drug and measure their average sleep time. They then randomly assign ten people their current medication (drug A) and ten people their new medication (drug B). The difference between hours of sleep with the sleep medications compared to hours of sleep under the placebo are recorded in the following table:

Drug A	0.7	-1.6	-1.2	0.1	-1.2	3.4	3.7	0.8	0.0	0.2
Drug B	1.9	0.8	1.1	0.1	-0.1	4.4	5.5	1.6	3.6	1.4

a Define the hypothesis set for this problem.

b Is this a one tailed or two tailed test?

c Find the p-value for the hypothesis, correct to three decimal places.

d Comment on your results if the test is conducted at the 5% significance level.

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6 A teacher is interested in whether there is a significant difference in the test scores of her female students in comparison to her male students. She assumes the test scores are normally distributed and decides to conduct a 2-sample t-test at a 10% level of significance to investigate.
The students' test scores are given in the following table:

Male	95	78	68	95	98	79	98	86	78	89	89	94
Female	100	100	95	90	95	98	100	100				

a Define the hypothesis set for this problem.

b Is this a one tailed or two tailed test?

c Find the p-value for the hypothesis, correct to two decimal places.

d Comment on your results.

e Explain the meaning of the 10% level of significance.

7 A company that specialises in helping people improve their presentation skills is interested in whether the reading levels of their native English speaking clients is significantly higher compared to their clients with English as a second language. They administer a reading test to their next 20 clients and record the scores in the table below.

Native speaker	94	85	79	91	86	77	68	91	90	89	88
Non-native speaker	96	87	86	78	77	91	85	80	90		

They assume the test scores are normally distributed and conduct a 2-sample t-test at a 5% level of significance to investigate.

a Define the hypothesis set for this problem.

b State the p-value for the hypothesis, correct to two decimal places, and comment on your findings.

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